Solution

(i) The electric field lines start at q_2 and end at q_1. In figure (a), q_2 is positive and q_1 is negative. The number of lines starting from q_2 is 18 and number of the lines ending at q_1 is 6. So q_2 has greater magnitude. The ratio of $\left|\frac{q_1}{q_2}\right| = \frac{N_1}{N_2} = \frac{6}{18} = \frac{1}{3}$. It implies that $|q_2| = 3|q_1|$

(ii) In figure (b), the number of field lines emanating from both positive charges are equal (N=18). So the charges are equal. At point A, the electric field lines are denser compared to the lines at point B. So the electric field at point A is greater in magnitude compared to the field at point B. Further, no electric field line passes through C, which implies that the resultant electric field at C due to these two charges is zero.

(iii) In the figure (c), the electric field lines start at q_1 and q_3 and end at q_2. This implies that q_1 and q_3 are positive charges. The ratio of the number of field lines is $\left|\frac{q_1}{q_2}\right| = \frac{8}{16} = \left|\frac{q_3}{q_2}\right| = \frac{1}{2}$, implying that q_1 and q_3 are half of the magnitude of q_2. So $q_1 = q_3 = +10$ nC.

1.4 ELECTRIC DIPOLE AND ITS PROPERTIES

1.4.1 Electric dipole

Two equal and opposite charges separated by a small distance constitute an electric dipole. In many molecules, the centres of positive and negative charge do not coincide. Such molecules behave as permanent dipoles. Examples: CO, water, ammonia, HCl etc.

Consider two equal and opposite point charges $(+q, -q)$ that are separated by a distance $2a$ as shown in Figure 1.14(a).
The electric dipole moment is defined as

$$\vec{p} = q\vec{r}_+ + (-q)\vec{r}_- \qquad (1.9)$$

where $\vec{r}_+$ is the position vector of $+q$ from the origin and $\vec{r}_-$ is the position vector of $-q$ from the origin. Then, from Figure 1.14 (a),

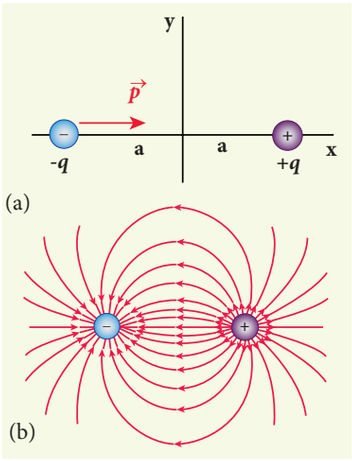


Figure 1.14 (a) Electric dipole (b) Electric field lines for the electric dipole

$$\vec{p} = qa\hat{i} - qa(-\hat{i}) = 2qa\hat{i} \qquad (1.10)$$

The electric dipole moment vector lies along the line joining two charges and is directed from $-q$ to $+q$. The SI unit of dipole moment is coulomb metre (Cm). The electric field lines for an electric dipole are shown in Figure 1.14 (b).

- For simplicity, the two charges are placed on the x-axis. Even if the two charges are placed on y or z-axis, dipole moment will point from $-q$ to $+q$.
- The magnitude of the electric dipole moment is equal to the product of the magnitude of one of the charges and the distance between them,

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